

## SIBOLGA PINANGSORI, Indonesia

WMO: 960730

Lat: 1.556N

Lon: 98.889E

Elev: 13

StdP: 101.17

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>11     | (b)<br>21.3 | (c)<br>21.8 | (d)<br>20.1                   | (e)<br>14.8 | (f)<br>25.6 | (g)<br>20.7 | (h)<br>15.4 | (i)<br>25.3 | (j)<br>5.8            | (k)<br>28.7 | (l)<br>5.0 | (m)<br>29.1 | (n)<br>0.1            | (o)<br>N/A | (p)<br>0.259 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 8.0                    | 33.2            | 25.9 | 32.7 | 25.8 | 32.2 | 25.8 | 27.6                | 31.1 | 27.2 | 30.7 | 26.9 | 30.4 | 2.6                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.6                            | 22.2 | 29.8 | 26.2 | 21.7 | 29.5 | 25.9 | 21.3 | 29.2 | 87.9          | 31.3 | 86.3 | 30.9 | 84.8 | 30.6 | 30.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 5.2           | (b) 4.4 | (c) 3.9 | (e) 18.9                   | (f) 34.6 | (g) 2.0            | (h) 0.7 | (i) 17.5                                           | (j) 35.1 | (k) 16.3   | (l) 35.5 | (m) 15.2   | (n) 35.9 | (o) 13.8   | (p) 36.3 |  |  |
|                   |         |         | DB                         | WB       | DB                 | WB      | DB                                                 | WB       | DB         | WB       | DB         | WB       | DB         | WB       |  |  |
|                   |         |         | 18.3                       | 28.8     | 1.7                | 0.8     | 17.1                                               | 29.4     | 16.2       | 29.9     | 15.2       | 30.4     | 14.0       | 31.0     |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.0   | 27.0  | 27.2  | 27.4  | 27.5  | 27.6  | 27.2  | 26.7  | 26.7  | 26.7  | 26.6  | 26.6  | 26.8  |
|                                                                        | DBStd       | 0.95   | 0.87  | 0.89  | 0.89  | 0.87  | 0.87  | 0.88  | 0.94  | 0.90  | 0.90  | 0.84  | 0.85  | 0.85  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD10.0     | 6206   | 527   | 483   | 539   | 525   | 546   | 517   | 519   | 517   | 502   | 516   | 497   | 520   |
|                                                                        | CDD18.3     | 3164   | 268   | 249   | 281   | 275   | 287   | 267   | 261   | 259   | 252   | 257   | 247   | 262   |
|                                                                        | CDH23.3     | 27848  | 2369  | 2278  | 2606  | 2572  | 2819  | 2489  | 2284  | 2162  | 2111  | 2081  | 1904  | 2171  |
|                                                                        | CDH26.7     | 9779   | 852   | 872   | 983   | 920   | 1070  | 942   | 832   | 744   | 688   | 636   | 547   | 694   |
| (14) Wind                                                              | WSAvg       | 0.6    | 0.5   | 0.6   | 0.6   | 0.5   | 0.6   | 0.6   | 0.6   | 0.6   | 0.6   | 0.6   | 0.5   | 0.5   |
| (15) Precipitation                                                     | PrecAvg     | 4076   | 277   | 280   | 395   | 383   | 283   | 230   | 308   | 264   | 373   | 439   | 446   | 410   |
|                                                                        | PrecMax     | 4971   | 507   | 487   | 677   | 571   | 705   | 483   | 599   | 557   | 651   | 854   | 790   | 747   |
|                                                                        | PrecMin     | 3253   | 47    | 57    | 82    | 202   | 73    | 68    | 105   | 61    | 193   | 122   | 221   | 187   |
|                                                                        | PrecStd     | 459    | 108   | 107   | 142   | 92    | 121   | 93    | 132   | 112   | 105   | 146   | 146   | 126   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 33.4  | 33.8  | 33.7  | 33.2  | 33.3  | 33.2  | 32.8  | 32.6  | 32.6  | 32.2  | 32.0  | 32.6  |
|                                                                        |             | MCWB   | 25.6  | 25.5  | 26.4  | 26.7  | 26.2  | 25.7  | 25.2  | 25.2  | 25.4  | 25.2  | 25.7  | 25.6  |
|                                                                        | 2%          | DB     | 32.3  | 32.7  | 32.8  | 32.4  | 32.6  | 32.5  | 32.1  | 31.9  | 31.9  | 31.5  | 31.2  | 31.7  |
|                                                                        |             | MCWB   | 25.6  | 25.7  | 26.2  | 26.5  | 26.1  | 25.7  | 25.3  | 25.3  | 25.4  | 25.3  | 25.6  | 25.7  |
|                                                                        | 5%          | DB     | 31.6  | 32.0  | 32.1  | 31.8  | 32.1  | 31.9  | 31.5  | 31.2  | 31.1  | 30.7  | 30.5  | 31.0  |
|                                                                        |             | MCWB   | 25.5  | 25.8  | 26.1  | 26.3  | 26.0  | 25.7  | 25.3  | 25.4  | 25.4  | 25.4  | 25.6  | 25.7  |
|                                                                        | 10%         | DB     | 30.8  | 31.3  | 31.2  | 31.0  | 31.4  | 31.2  | 30.7  | 30.4  | 30.3  | 30.0  | 29.7  | 30.2  |
|                                                                        |             | MCWB   | 25.5  | 25.7  | 25.9  | 26.2  | 26.0  | 25.7  | 25.4  | 25.4  | 25.4  | 25.4  | 25.5  | 25.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.4  | 27.9  | 27.9  | 28.0  | 27.7  | 27.5  | 27.2  | 27.1  | 27.0  | 27.0  | 27.2  | 27.2  |
|                                                                        |             | MCDB   | 31.0  | 31.3  | 31.7  | 31.5  | 31.2  | 31.2  | 30.6  | 30.4  | 30.3  | 30.2  | 30.2  | 30.5  |
|                                                                        | 2%          | WB     | 26.7  | 27.2  | 27.2  | 27.4  | 27.2  | 26.9  | 26.6  | 26.4  | 26.4  | 26.4  | 26.6  | 26.6  |
|                                                                        |             | MCDB   | 30.3  | 30.8  | 31.0  | 30.9  | 30.7  | 30.5  | 30.1  | 29.9  | 29.8  | 29.6  | 29.6  | 29.9  |
|                                                                        | 5%          | WB     | 26.2  | 26.6  | 26.7  | 27.0  | 26.8  | 26.5  | 26.2  | 26.0  | 26.0  | 26.0  | 26.1  | 26.2  |
|                                                                        |             | MCDB   | 29.9  | 30.3  | 30.5  | 30.5  | 30.3  | 30.1  | 29.7  | 29.7  | 29.5  | 29.2  | 29.2  | 29.5  |
|                                                                        | 10%         | WB     | 25.8  | 26.1  | 26.3  | 26.5  | 26.5  | 26.1  | 25.7  | 25.6  | 25.6  | 25.6  | 25.7  | 25.8  |
|                                                                        |             | MCDB   | 29.4  | 29.9  | 30.0  | 29.9  | 30.0  | 29.8  | 29.2  | 29.2  | 29.0  | 28.7  | 28.6  | 29.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.2   | 8.6   | 8.3   | 7.7   | 8.0   | 8.2   | 8.2   | 7.8   | 7.5   | 7.1   | 6.8   | 7.3   |
|                                                                        |             | MCDBR  | 9.3   | 9.6   | 9.3   | 8.6   | 8.9   | 9.3   | 9.4   | 9.0   | 8.7   | 8.3   | 7.9   | 8.5   |
|                                                                        |             | MCWBR  | 3.8   | 3.9   | 3.8   | 3.5   | 3.5   | 3.7   | 3.8   | 3.7   | 3.4   | 3.3   | 3.3   | 3.5   |
|                                                                        | 5% WB       | MCDBR  | 8.4   | 8.7   | 8.6   | 8.0   | 8.2   | 8.5   | 8.4   | 8.3   | 7.9   | 7.4   | 7.2   | 7.7   |
|                                                                        |             | MCWBR  | 3.8   | 4.2   | 4.0   | 3.7   | 3.7   | 3.9   | 4.0   | 3.8   | 3.4   | 3.3   | 3.4   | 3.6   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.458 | 0.535 | 0.506 | 0.446 | 0.434 | 0.440 | 0.432 | 0.462 | 0.467 | 0.501 | 0.442 | 0.423 |
|                                                                        | taud        |        | 2.332 | 2.087 | 2.196 | 2.378 | 2.424 | 2.412 | 2.430 | 2.328 | 2.311 | 2.200 | 2.387 | 2.442 |
|                                                                        | Ebn at Noon |        | 871   | 810   | 831   | 863   | 852   | 834   | 846   | 837   | 851   | 829   | 881   | 899   |
|                                                                        | Edh at Noon |        | 132   | 171   | 153   | 125   | 115   | 114   | 113   | 129   | 135   | 151   | 124   | 117   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 4.99  | 5.29  | 5.39  | 5.20  | 5.22  | 5.05  | 5.00  | 4.94  | 4.82  | 4.76  | 4.66  | 4.73  |
|                                                                        | RadStd      |        | 0.22  | 0.30  | 0.29  | 0.21  | 0.24  | 0.20  | 0.26  | 0.25  | 0.37  | 0.58  | 0.19  | 0.21  |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | N/A     | N/A    | +0.38   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | +0.38   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page